Definitions glossary – with where to cite each one

Definitions in plain words plus the source to cite. **Check the wording against the book and write it in your own words.** "S x.y" = Strogatz, *Nonlinear Dynamics and Chaos* (section numbers from the 2nd ed.; chapters are the same in the 3rd). "Murray" = *Mathematical Biology II*, ch. 2.

How to use a definition in the report

Your preamble already has a `definition` environment. Cite the book with a section or page in square brackets:

```
\begin{definition}[Fixed point]\label{def:fixed-point}
A point $\mathbf{x}^*$ with $\mathbf{f}(\mathbf{x}^*) = \mathbf{0}$ is a fixed point of
$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ \parencite[Sec.~2.2]{strogatz2024}.
\end{definition}

... inline: following \textcite[Sec.~8.2]{strogatz2024}, a Hopf bifurcation occurs when ...
... later: by \cref{def:fixed-point}, ...
```

Rule of thumb: use a `definition` box only for the 3–5 terms your argument depends on (fixed point, linear stability, bifurcation, Turing instability). Define everything else in a sentence with a citation.

Term	Definition (plain words)	Cite
Dynamical systems basics		
Dynamical system	A rule for how a state changes in time; here a system of ODEs $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ (continuous time).	S 1.2
Phase space / phase portrait	The space of all possible states; the phase portrait is the picture of trajectories in it.	S 2.1, 6.1
Nonlinear	$\mathbf{f}$ is not linear in $\mathbf{x}$ (products like UV^2, powers like x^3). Superposition fails.	S 1.2
Fixed point (equilibrium, steady state)	$\mathbf{x}^*$ with $\mathbf{f}(\mathbf{x}^*) = \mathbf{0}$: if you start there, you stay there.	S 2.2, 6.3
Homogeneous (uniform) steady state	For a PDE: a steady state that is the same at every point in space, so all diffusion terms vanish.	Murray ch. 2
Attracting	Every trajectory starting close enough to $\mathbf{x}^*$ approaches it as $t \to \infty$.	S 5.1
Lyapunov stable	Trajectories starting close enough stay close for all time (need not approach).	S 5.1
(Asymptotically) stable	Both attracting and Lyapunov stable.	S 5.1
Linearisation	Approximate $\mathbf{f}$ near $\mathbf{x}^*$ by its first-order Taylor expansion: $\dot{\boldsymbol{\eta}} = J\boldsymbol{\eta}$, $\boldsymbol{\eta} = \mathbf{x} - \mathbf{x}^*$.	S 2.4, 6.3
Jacobian	Matrix of first partial derivatives $J_{ij} = \partial f_i / \partial x_j$, evaluated at the fixed point.	S 6.3
Linear stability	Stability decided by the eigenvalues of J: stable if all have negative real part.	S 2.4, 6.3
Hyperbolic fixed point	No eigenvalue of J has zero real part.	S 6.3
Hartman–Grobman theorem	Near a hyperbolic fixed point, the nonlinear phase portrait is topologically equivalent to the linearised one.	S 6.3 (Perko for proof)
Trace–determinant classification	With $p = \operatorname{tr} J$, $q = \det J$: $q < 0$ saddle; $q > 0$, $p^2 > 4q$ node; $q > 0$, $p^2 < 4q$ spiral; $p = 0$, $q > 0$ centre. Stable iff $p < 0,\ q > 0$.	S 5.2
Saddle / node / spiral / centre	Saddle: one growing and one decaying direction. Node: all trajectories approach (or leave) without circling. Spiral: approach/leave while rotating (complex eigenvalues). Centre: closed orbits around the point.	S 5.2

Term	Definition (plain words)	Cite
Bistability	Two stable states exist for the same parameters; the initial condition decides which is reached.	S 3.7, 6.4
Basin of attraction	The set of initial conditions that end up at a given stable state.	S 6.2, 6.4
Conservative system	Has a conserved quantity $E(\mathbf{x})$ (energy) constant along trajectories. For $\ddot{x} = g(x)$: $E = \frac{1}{2}\dot{x}^2 + E_P(x)$ with $E_P = -\int g\,dx$.	S 6.5
Limit cycle	An isolated closed trajectory; nearby trajectories spiral towards it (stable) or away (unstable).	S 7.0
Poincaré–Bendixson theorem	In the plane, a trajectory confined to a closed bounded region with no fixed points must approach a closed orbit. Hence no chaos in 2D autonomous systems.	S 7.3

Bifurcations

Bifurcation	A qualitative change in the dynamics (fixed points created/destroyed or changing stability) as a parameter passes a critical value.	S 3.0
Bifurcation point / curve	The critical parameter value; with two parameters (e.g. F, k) these form curves in parameter space.	S 3.0, 8.1
Saddle-node bifurcation	Two fixed points collide and annihilate. Normal form $\dot{x} = \gamma - x^2$. In 2D: a real eigenvalue passes through 0 ($\det J = 0$).	S 3.1, 8.1
Transcritical bifurcation	Two fixed points pass through each other and exchange stability. $\dot{x} = \gamma x - x^2$.	S 3.2
Pitchfork bifurcation	One fixed point becomes unstable and two symmetric ones appear (supercritical) or vice versa. $\dot{x} = \gamma x - x^3$.	S 3.4
Hopf bifurcation	A complex-conjugate pair of eigenvalues crosses the imaginary axis ($\operatorname{tr} J = 0$, $\det J > 0$): a fixed point loses stability and a limit cycle is born (supercritical) or destroyed (subcritical).	S 8.2
Normal form	Simplest equation showing a given bifurcation; any system near that bifurcation looks like it.	S 3.1
Parameter space	The space of parameter values (here the (k, F) plane), divided by bifurcation curves into regions of qualitatively different behaviour.	S 3.7

Reaction–diffusion and pattern formation

Reaction–diffusion system	PDEs $\partial_t \mathbf{u} = \mathbf{f}(\mathbf{u}) + D\nabla^2\mathbf{u}$: local reactions plus spreading by diffusion.	Murray ch. 2; `turing1952`
Diffusion / Laplacian	$\nabla^2 u$ compares u at a point with its surroundings; diffusion moves material from high to low concentration and smooths differences.	Murray ch. 2
Diffusion-driven (Turing) instability	A homogeneous steady state that is stable without diffusion becomes unstable to spatially varying perturbations when diffusion is added.	Murray ch. 2; `turing1952`
Turing pattern	The steady spatial pattern (spots/stripes) that grows from a Turing instability; its wavelength is set by the system, not by the domain or initial condition.	Murray ch. 2; `kondo2010`
Wavenumber / wavelength	A perturbation $\cos(qx)$ has wavenumber q and wavelength $2\pi/q$.	Murray ch. 2
Dispersion relation	Growth rate $\lambda(q)$ of each wavenumber, from the eigenvalues of $J - q^2 D$. Instability if $\operatorname{Re}\lambda(q) > 0$ for some $q \neq 0$.	Murray ch. 2
Critical wavenumber q_c	Wavenumber at which instability first appears (minimum of $\det(J - q^2 D)$).	Murray ch. 2
Activator / inhibitor	Activator: promotes its own production ($\partial f/\partial u > 0$). Inhibitor: suppresses the activator. Turing needs "short-range activation, long-range inhibition" (inhibitor diffuses faster).	Murray ch. 2; `kondo2010`
Autocatalysis	A chemical speeds up its own production; in Gray–Scott, $U + 2V \to 3V$.	`grayscott1984`
Gray–Scott model	$\partial_t U = D_u\nabla^2 U - UV^2 + F(1-U)$, $\partial_t V = D_v\nabla^2 V + UV^2 - (F+k)V$; F = feed rate, k = rate of $V \to P$ ("kill").	`grayscott1984`, `pearson1993`

Term	Definition (plain words)	Cite
Finite-amplitude perturbation	A disturbance too large for linear theory to describe. Pearson's patterns arise this way from a linearly stable state.	`pearson1993`
Self-replicating spots	Spots that grow and divide in two (Pearson's patterns in part of the (k, F) plane).	`pearson1993`
Numerics (for Section 4)		
Finite differences	Replace derivatives by differences on a grid; the 5-point Laplacian is $(u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1} - 4u_{i,j})/\Delta x^2$.	any numerical PDE text; `pearson1993`
Explicit (forward) Euler	$u^{n+1} = u^n + \Delta t\,(\text{RHS at } u^n)$. Simple; stable for diffusion only if $4D\Delta t/\Delta x^2 \leq 1$ (2D).	standard; state the condition you used
Periodic boundary conditions	The grid wraps around (left edge neighbours right edge); gives tileable textures.	`pearson1993`
Graphics		
Texture mapping	Wrapping a 2D image onto a 3D surface by assigning texture coordinates to its points.	`turk1991`
Procedural texture	A texture computed by a function/algorithm from a few parameters rather than stored as an image.	`ebert2003`, `lagae2010`
Procedural noise function	A function of position giving smooth, controllable pseudo-random variation (Perlin, simplex, Worley, Gabor).	`lagae2010`, `perlin1985`
Fractal Brownian motion (fBm) / turbulence	Sum of noise at several scales (octaves) with decreasing amplitude, giving detail at every scale.	`ebert2003`, `lagae2010`
Example-based texture synthesis	Growing a large texture that resembles a small sample image.	`wei2009`, `efros1999`
Procedural content generation (PCG)	Algorithmic creation of game content (textures, levels, terrain, items).	`hendrikx2013`, `shaker2016`

Not to confuse (your own earlier notes): *chaos* = sensitive dependence on initial conditions in a bounded, aperiodic system (S 9.3) – Turing patterns are *not* chaos, and noise-based textures are *not* chaos. Use "nonlinear dynamics".